

ASSESSMENT TOOL – 2

20/20

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO1	Assessment	Tool – Concept Mapping
Weightage		Total Marks	

Student Name	Cindy R	Reg. No.	192412571
Section / Batch	ECE-2024	Date of Submission	
Faculty Incharge	Dr R S Selvan	Assessment	Individual Submission

Total Marks		Obtained Marks	
--------------------	--	-----------------------	--

Code of Conduct:

I Cindy R/192412571 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



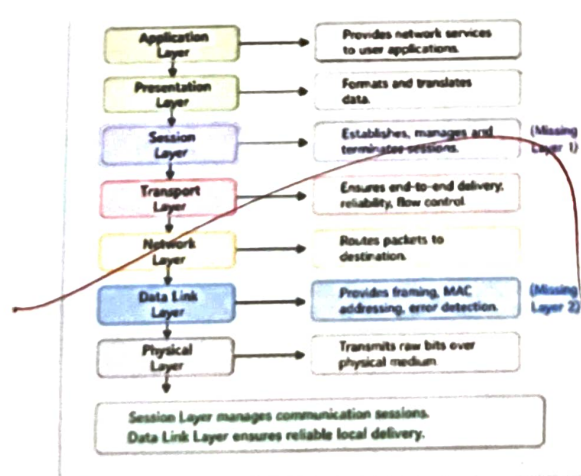
Q1. OSI Layers and Reliable Communication

Concept Map

Source Concept	Linking Relationship	Target Concept
Application	passes user data to	Presentation
Presentation	formats data for	Session Layer
Session Layer	establishes and manages	Transport
Transport	provides end-to-end delivery to	Network
Network	forwards packets to	Data Link Layer
Data Link Layer	sends frames through	Physical

Analysis

- **Session Layer** establishes, maintains, and terminates communication sessions between devices.
- **Data Link Layer** provides framing, MAC addressing, and error detection for reliable local communication.
- Together, these layers ensure organized communication, reliable data transfer, and proper delivery across the network.



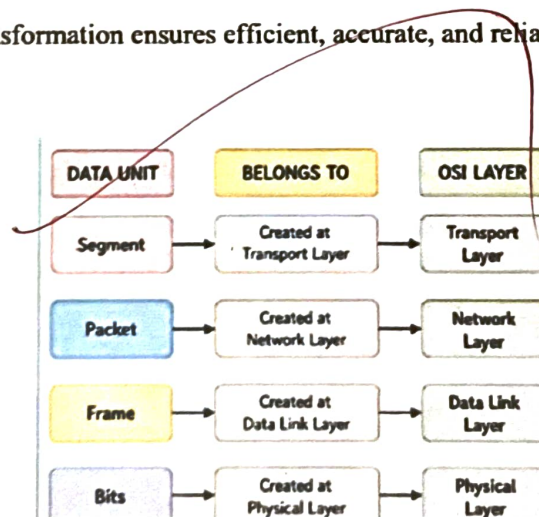
Q2. Data Units and OSI Layers

Concept Map

Data Unit	Belongs to	OSI Layer
Segment	→	Transport Layer
Packet	→	Network Layer
Frame	→	Data Link Layer
Bits	→	Physical Layer

Analysis

- The **Transport Layer** divides data into **segments**.
- The **Network Layer** adds IP information to create **packets**.
- The **Data Link Layer** adds MAC information to form **frames**.
- The **Physical Layer** converts frames into **bits** for transmission.
- This transformation ensures efficient, accurate, and reliable communication.



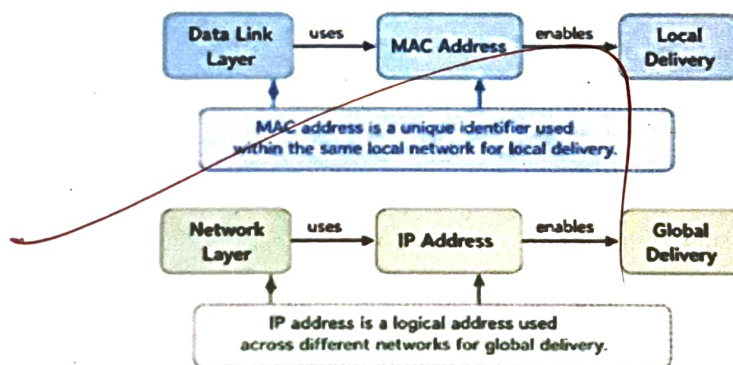
Q3. MAC Address and IP Address

Concept Map

Source Concept	Linking Relationship	Target Concept
Data Link Layer	uses	MAC Address
MAC Address	enables	Local Delivery
Network Layer	uses	IP Address
IP Address	enables	Global Delivery

Analysis

- The **MAC address** identifies devices within the same local network.
- The **IP address** identifies devices across different networks.
- Together, MAC and IP addressing ensure data reaches the correct destination locally and globally.



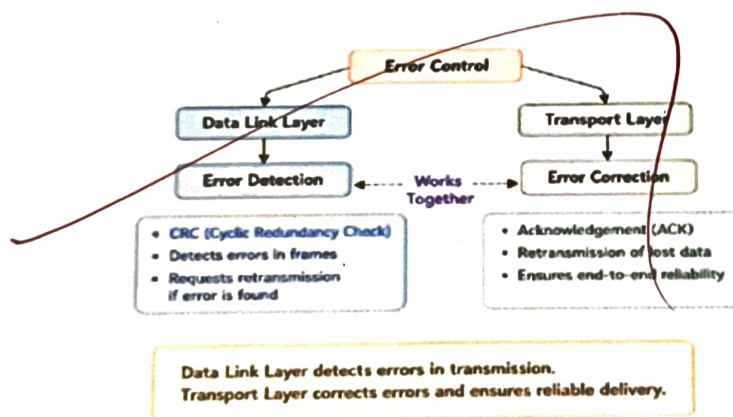
Q4. Error Control

Concept Map

Source Concept	Linking Relationship	Target Concept
Error Control	includes	Data Link Layer
Data Link Layer	performs	Error Detection
Error Control	includes	Transport Layer
Transport Layer	performs	Error Correction

Analysis

- The **Data Link Layer** detects transmission errors using methods like CRC.
- The **Transport Layer** ensures reliable delivery through acknowledgments and retransmissions.
- Together, these layers reduce data loss and improve communication reliability.



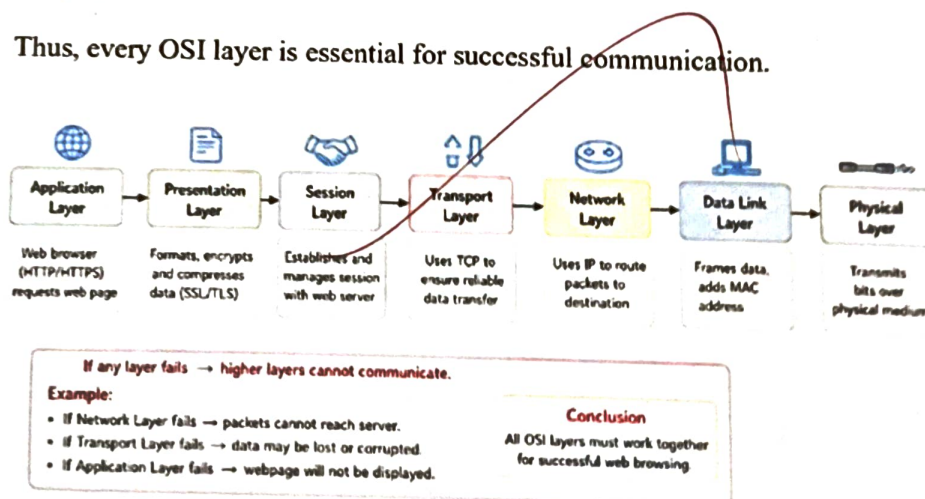
Q5. Web Browsing Using OSI Layers

Concept Map

Source Concept	Linking Relationship	Target Concept
Web Browser	uses	Application Layer
Application Layer	formats data through	Presentation Layer
Presentation Layer	establishes session with	Session Layer
Session Layer	transfers data via	Transport Layer
Transport Layer	routes through	Network Layer
Network Layer	frames data using	Data Link Layer
Data Link Layer	transmits bits through	Physical Layer

Analysis

- During web browsing, all OSI layers work together to deliver web pages.
- If one layer fails (e.g., Network Layer), higher layers cannot communicate, causing the webpage to fail to load.
- Thus, every OSI layer is essential for successful communication.



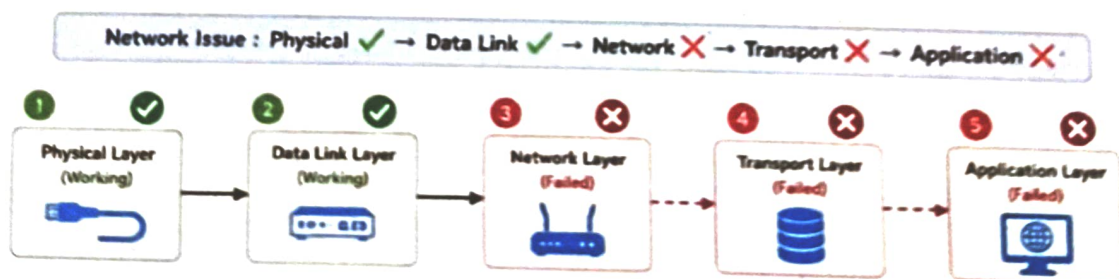
Q6. Troubleshooting Using OSI Layers

Concept Map

Layer	Status
Physical	✓ Working
Data Link	✓ Working
Network	✗ Failed
Transport	✗ Failed
Application	✗ Failed

Analysis

- Since the **Physical** and **Data Link** layers are functioning, the physical connection is intact.
- Failure at the **Network Layer** prevents packet routing.
- As a result, the **Transport** and **Application** layers also fail because they depend on the Network Layer.
- The concept map helps troubleshoot by identifying the exact layer where communication breaks, making fault diagnosis faster and easier.



22/11/20
20/11/20